

Heena Khan

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EDUCATION

Stanford University

Expected Graduation: June 2027 (B.S.)

B.S. Computer Science (Artificial Intelligence) | Minor: Biology

Relevant Coursework: Operating Systems, Compilers, Data Structures & Algorithms, Databases, Probability & Statistics, Machine Learning, Discrete Mathematics, Cloud Computing, Computer Vision

EXPERIENCE

Software Engineering Intern — Data Infrastructure & Full Stack

Santa Clara, CA | June 2026 – October 2026

Roche / Genentech

- Built domain-specific AI agent translating natural language queries into SQL over a 25+ table production PostgreSQL database; designed ground-truth Q&A evaluation framework to measure agent accuracy end-to-end.
- Architected relational PostgreSQL database schemas with optimized multi-column indexing to store and query complex PCR and NGS assay records across Roche's diagnostic test portfolio.
- Engineered full-stack Django and React web applications featuring interactive data visualizations to graph real-time PCR amplification curves and genomic signals.
- Built containerized Python and SQL ETL pipelines using Docker, deploying automated CI/CD schema-validation workflows to protect against data drift across high-performance Linux cluster storage.

AI Research Engineering Intern

Palo Alto, CA | June 2025 – September 2025

Neurotrack Technologies

- Engineered high-throughput data curation and feature pipelines in Python/SQL, ingesting and normalizing 10,000+ multimodal assessment records across 9 heterogeneous input schemas.
- Designed automated validation and noise-filtering algorithms to detect schema drift and artifact corruption, increasing input dataset reliability for downstream training pipelines.
- Trained and evaluated linear regression and parametric curve-fitting models in Python to map score distribution trajectories, using statistical evaluation metrics to prune 6 redundant assessment variants.

Stanford Tech Fellowship

Stanford, CA | June 2023 – September 2024

Pull for Progress

- Engineered geospatial ML models in Python using multidimensional health metrics to predict regional risk profiles across 50,000+ healthcare data points.
- Applied statistical modeling and spatial feature extraction to optimize healthcare worker deployment in resource-limited settings.

PROJECTS

AMANI — AI-Enabled Health Mapping System | Stanford Healthcare Design Challenge Finalist

React Native, TensorFlow Lite, PostgreSQL, REST APIs, Computer Vision

- Led development of offline-first mobile app integrating on-device computer vision to classify WASH hazards in low-connectivity environments; quantized TFLite models to run on low-RAM Android devices.
- Designed privacy-preserving system with anonymous reporter IDs, AES-256 encrypted sync payloads, and human-in-the-loop verification — AI suggests classification, frontline staff confirm before dispatch.
- Built offline sync queue with exponential backoff and geo-proximity deduplication to preserve data integrity across intermittent connectivity.

Khaista Boutique — Full-Stack E-Commerce Platform

TypeScript, React, Node.js, PostgreSQL, Drizzle ORM, Stripe, Vite

- Built end-to-end e-commerce platform with React + TypeScript frontend, Node.js/Express REST API, and PostgreSQL with Drizzle ORM for Afghan women artisans selling handmade goods globally.
- Integrated Stripe payments, server-side input validation with Zod, and automated CI/CD via GitHub Actions with environment-specific deployments.

ModestFilter — Chrome Extension (Manifest V3)

TypeScript, React, Chrome APIs, MutationObserver, chrome.storage

- Built Manifest V3 Chrome extension in TypeScript and React with real-time keyword filtering across 7+ e-commerce sites, MutationObserver-based DOM handling for infinite scroll, and chrome.storage sync for persistent preferences.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java, C++

Frameworks & Libraries: React, Node.js, Django, Vite, Pandas, NumPy, TensorFlow Lite

AI / ML: LLMs, Prompt Engineering, NL-to-SQL Agents, Computer Vision, Embeddings, Statistical Modeling

Tools & Systems: PostgreSQL, Docker, Git, Linux, Bash, CI/CD, REST APIs, Chrome APIs